



Computer Organization and Assembly Language

Lab # 1

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TASK:

1. Difference between Compiler, Assembler and Interpreter?

Compiler:

A compiler is a program that translates the entire high-level language program (like C, C++) into machine code at once. After compilation, an executable file is created, and the program runs faster. If there is an error, the compiler shows all errors after compiling the whole program.

Assembler:

An assembler converts assembly language (low-level language) into machine language. It works very close to the hardware and translates instructions almost one-to-one. Assembly language is easier than machine code but harder than high-level languages.

Interpreter:

An interpreter translates and executes the program line by line. It does not create an executable file, so the program runs slower. If an error occurs, the interpreter stops immediately at that line.

Examples: Python and JavaScript.

Comparison Table:

Feature	Compiler	Assembler	Interpreter
Language	High-level	Assembly	High-level
Translation	Whole program	One-to-one	Line by line
Output file	Yes	Yes	No
Speed	Fast	Very fast	Slow

THE END